Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 31)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

0 1 . 1 Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

0 1 . 2 Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

0 1 . 3 Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

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0 2 Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 2 . 1 Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

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0 2 . 2 Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

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0 2 . 3 Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

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0 2 . 4 Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

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0 3 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 3 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 3 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 4 A national conservation body must decide how to divide a limited budget between species.

0 4 . 1 Describe **three** criteria the IUCN uses when deciding the Red List category of a species.

[3 marks]

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0 4 . 2 A population of lemurs has fallen from 12 000 to 2 100 mature individuals. Calculate the percentage decline.

[2 marks]

Answer ..... %

0 4 . 3 Explain **three** reasons, other than its Red List category, why one species may be given a higher conservation priority than another.

[3 marks]

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0 4 . 4 Explain the difference between CITES Appendix I and Appendix II, and explain why controlling trade can protect a species more effectively than habitat protection alone.

[3 marks]

0 4 . 5 Describe **four** ways that UK legislation and designated areas protect habitats.

[4 marks]

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0 5 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 5 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 5 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 6 . 1 Explain the effects of deforestation on soil.

[4 marks]

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0 6 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

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0 6 . 3 Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

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0 7 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 7 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 8 Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 8 . 1 Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer ..... %

0 8 . 2 Explain why so little of the solar energy is stored in the crop.

[3 marks]

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0 8 . 3 Explain why few food chains have more than four trophic levels.

[2 marks]

0 8 . 4 Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

0 8 . 5 Explain how pigs can increase the efficiency of a food production system.

[2 marks]

10

0 9 Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 9 . 1 Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

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0 9 . 2 Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

0 9 . 3 Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

0 9 . 4 Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

10

1 0 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

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1 0 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

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1 0 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

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1 0 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS